



Unit 11: Project Management Software

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1. Project Management Software

Introduction to the Unit

In this unit, we will look at the advantages and common features of the various types of *Project Management Software*.

We will focus on tracing the evolution of *Project Management Software*, its *Advantages*, the *Common Features* shared by various software packages and will conclude the discussion with an illustration of a common software management package known as the *Gantt chart*.

Objectives

Upon completion of this unit, you will be able to:

- Describe the advantages of using project management software
- Identify the common features in most types of project management software
- Explain how a Gantt chart is used



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2. Advantages of Using Project Management Software
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Introduction

In the last segment, we learned about the reasons why most projects fail and about the warning signs that can help us mitigate the risk of failure by taking corrective action in time. In this segment, we will focus on the advantages and common features of the various types of project management software.

In the early days, project management software was used only on mainframe computers. The software had limited capability and was relatively less user friendly. Its application was also limited. With the growth of business, more and more projects of varying complexity have come up, all of which need to be managed.

This has forced the qualitative and quantitative growth of software. Gradually, more and more features and functions have been added into the software.

Today, we have a large number of PC-based project management software packages available. These are more flexible in accommodating user requirements, capable of handling increased functions and larger size projects with speed and accuracy and are available at a reasonable cost.

We will also discuss other aspects of the project management software such as the need for software, an overview of common features available in most packages, how to get the best of these and general information about them.

Advantages of Using Project Management Software

Following are the key advantages of using project management software:

Increased speed and accuracy with reduced effort

For a large project, manually carrying out activities such as drawing a network, carrying out time analysis, reporting project progress, generating various types of reports, updating network and maintaining records is quite time and effort consuming. Accuracy level is also below par. Use of the



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software package greatly reduces the time and effort needed for these activities and at the same time enhances accuracy.

Ability to carry out special functions

The software has the ability to carry out some special functions such as resource scheduling, what-if experiment and export and import of data with ease and within reasonable time. Manually carrying out these functions is either extremely difficult or not feasible.

Affordability

The price of PC-based software is under \$500, which is affordable for an organisation.

Easy to use

Over the years, the project management software packages have become easy to use. This package can now be handled with minimum training.

Provision for maintenance of records

A project generates a lot of data, reports, documents, etc. Archiving and retrieving them manually are time and effort consuming. The software package can handle these functions with relatively less time, effort and cost.

Features Common to Most Types of Project Management Software

Before knowing the practical use of project management software, we should study some of the common features available in most types of project management software. Let us have an overview of the generic features associated with various types of project management software packages available today.

Data entry features

Project data and calendar: A project start date is specified. A calendar can be used to define the working days and hours for each individual resource on a project. The calendar is used for calculating the schedule for the project. Most systems provide a default for the standard working period, such as Monday to Friday from 8:00 am to 5:00 am, with an hour's break for lunch. The calendar can be modified for each resource.

For example, work hours can be modified, company holidays can be entered as non-working days and various shifts can also be entered.

Human resources: Suppose a particular activity needs two unskilled resources and one skilled resource to complete the task. These two resources may be entered separately and will appear as 2L and 1S on activity description in the network.



Type of resource	Amount of resource available (No.)	Period	
		From	To
Skilled person	10		
Unskilled person	100		

Figure 33: Types of Resources and Amount Available

Labour cost: One of the many ways to specify labour cost is as:

- Skilled worker – USD 2.0
- Unskilled worker – USD 1.2

Human resources available: All software requires periods and amount of resources that are available for the project.

Cost of construction materials: Materials needed for each task and their estimated costs for the project may be as shown in figure 34.

Activity ID	Activity Description	Type of Materials	Estimated Cost (\$)
001	A	Wood	200
002	B	Pipe	300
...	...	...	...
00n	N	Flooring compound	500

Figure 34: Materials Needed and the Estimated Cost

Activity identifier: Each activity of the project is assigned a code or identifier.

Activity description: Each activity has a description. The number of characters should be within the number of characters specified by the software for the activity name field.

Precedence relationship: There are various options to show the linkage between two consecutive activities in a network. One common option is Finish-Start type. It means that the succeeding activity can start only if all preceding activities to it have been completed. This option (Finish-Start type) is the default option in all project management software today for linking two activities.

Data entry error detection: All good management software contain error-detection routines that identify and report errors. However, there are certain errors, which cannot be detected by a software. These include:

- an incorrect activity duration,
- an incorrect activity name,
- an Incorrect precedence logic,
- an Incorrect cost data, and



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- a wrong resource entered.

These errors have to be corrected manually by listing the activity records.

Graphics: For a project involving a large number of activities, drawing a correct network, manually, takes a lot of time and effort.

One of the important features of project management software is its ability to generate a variety of charts including network diagram, activity-linked Gantt chart, and Gantt chart quickly. Further, changes in base line plan are quite easy.

Time analysis: If there is an unlimited or flexible resource or if the resource can be outsourced, the network may be prepared at the earliest start time of activities. In the real world, many projects are managed on this basis.

Project Management packages carry out time analysis, which includes calculation of early start, early finish, late start, late finish, free slack and total slack with ease. Manual time analysis can be tedious.

Resource scheduling: Resource scheduling problems are of two types:

- Resource levelling where unlimited and flexible resources are available.
- Resource allocation problems where resources are limited.

In resource levelling, activities are scheduled to minimise the variation in the level of resource deployment.

The resource allocation problem is concerned with scheduling activities in such a way so as to find the shortest project schedule.

The project management software uses heuristics to solve both types of problems. For a moderate size problem involving 100 activities, it becomes extremely difficult to carry on resource scheduling. PM software is an invaluable tool to deal with this problem.

Output reports: Most project management software packages have extensive report generation capabilities. They can generate a range of reports in various forms (graphical, tabular or textual). The reports are standard or customised.

The content of each report for a recipient is based on a 'need-to-know basis' and is presented in a particular order. These requirements are met by two features of the software, namely filtering (editing) and sorting.

Filtering enables the user to select only certain data that meet some specified criteria. Sorting, on the other hand, allows the user to view information in a desired order such as pay rates from highest to lowest or in alphabetical order. Most software allows multiple level of sorting (for example, by year and then by dates).

The reports generated by the software include:

Project schedule: Network (based on AON systems), linked Gantt chart, Gantt chart.

- Work-to list



Cost related report: Budgeted versus. actual cost (daily and cumulative).

- Resources utilisation report

Progress report: Overall project, milestone chart and critical path.

- Chart showing responsibility of department/function to carry out particular activities.
- Progress summary report.

Updating: Updating is the process of producing a fresh set of schedule and other reports to take account of one or more of the following:

- A change to the project parameter, an unexpected increase or decrease in the resource available, changed cost rate or new target dates.
- A change in network logic due to change in project scope, design change, etc.
- To have a new schedule that takes into account the progress made to date.

Importing/exporting: The process of bringing information into the project management software from other application such as word processing, spread sheet, etc. is called importing. Similarly, sending information from the software to other applications is called exporting.

The exporting and importing obviates typing/data entry effort and eliminates the possibility of associated error.

Project monitoring and tracking: Tracking the progress for schedule and cost is an important aspect of project management. Most project management software packages permit the users to define a baseline plan and compare the actual progress with respect to those in the baseline plan.

What if analysis: This is a useful feature of the project management software. It permits to know the effect of changes in the project variable (people, cost, and change in scope) on project objective. This analysis helps the *Project Manager* in taking an appropriate decision.

Understanding a Gantt chart

Gantt chart is a tool for examining and scheduling complex projects. It facilitates the planning of tasks that need to be completed and provides a basis for scheduling when these tasks will be carried out. When a project is under way, Gantt Charts help to monitor whether the project is on schedule or not. If it is not, a Gantt chart helps to determine the corrective action required to put the project back on schedule.

Now, let us understand how Gantt chart actually works:

Consider the following illustration of some features of MS project on a sample project:

A sample project is described by the following data depicted in Figure 34.



Activity	Duration (Days)	Immediate Precedence	Manpower Required	Material Cost (₹)
A	2	-	2	100 (2 units, 200%)
B	4	A	4	50 (1 unit, 100%)
C	5	A	4	50 (1 unit, 100%)
D	6	A	4	150 (3 unit, 150%)
E	8	B,C	6	100 (2 unit, 100%)
F	2	D	4	200 (4 unit, 400%)
G	7	E,F	4	50 (1 unit, 100%)
H	8	D	4	50 (1 unit, 100%)
I	4	G,H	4	50 (1 unit, 100%)

Figure 34: Sample Project Data

With maximum labour available as unlimited and 8 persons, prepare the following, using MS project:

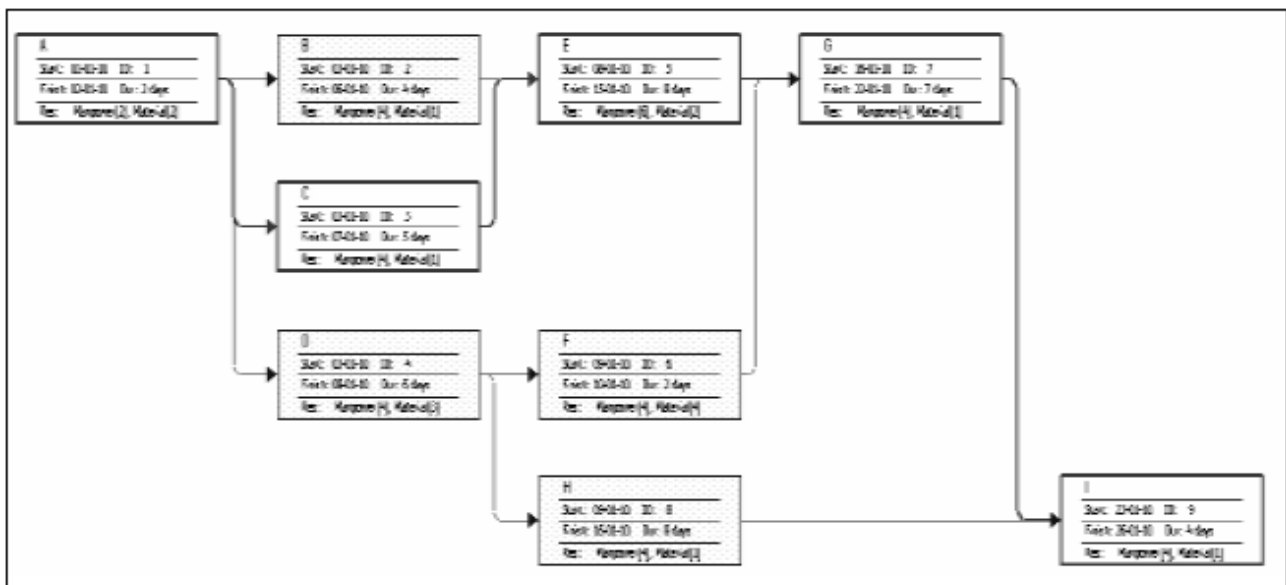
1. Network diagram
2. With unlimited resources:
 - Gantt chart
 - Time analysis
 - Resource graph
3. With limited resources (8 persons):
 - Gantt chart
 - Time analysis
 - Resource graph
4. Tracking Gantt chart
5. Network diagram showing work status

Solutions

Using MS Project, solutions for the above questions pertaining to the sample project were obtained and these are depicted in Figure 35 to Figure 42. Based on these outputs, some observations are given below.



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ID	Task Name	Duration	Early Start	Early Finish	Late Start	Late Finish
1	A	2 days	01-01-10	02-01-10	01-01-10	02-01-10
2	B	4 days	03-01-10	06-01-10	04-01-10	07-01-10
3	C	5 days	03-01-10	07-01-10	03-01-10	07-01-10
4	D	6 days	03-01-10	08-01-10	08-01-10	13-01-10
5	E	8 days	08-01-10	15-01-10	08-01-10	15-01-10
6	F	2 days	09-01-10	10-01-10	14-01-10	15-01-10
7	G	7 days	16-01-10	22-01-10	16-01-10	22-01-10
8	H	8 days	09-01-10	16-01-10	19-01-10	26-01-10
9	I	4 days	23-01-10	26-01-10	23-01-10	26-01-10

Figure 35: Network Diagram

1. With unlimited resources (14 men or more), the project will take 26 days to complete.
2. With limited resources of 8 persons, the project duration would be 33 days.
3. We get early start, early finish, late start, late finish and total slacks of each activity.
4. Tracking chart and network diagram may be used to track the progress of work.

Above are only a few application of MS project. Many more applications including project tracking, what-if analysis and others can be done with the help of the package.

Assumption:

- (i) 7 day working for manpower
- (ii) No constraint on manpower

With Unlimited Manpower:



ID	Task Name	Duration	Start	Finish	Predecessors
1	A	2 days	01-01-10	02-01-10	
2	B	4 days	03-01-10	06-01-10	1
3	C	5 days	03-01-10	07-01-10	1
4	D	6 days	03-01-10	08-01-10	1
5	E	8 days	08-01-10	15-01-10	2,3
6	F	2 days	09-01-10	10-01-10	4
7	G	7 days	16-01-10	22-01-10	5,6
8	H	8 days	09-01-10	16-01-10	4
9	I	4 days	23-01-10	26-01-10	7,8

a. Gantt Chart

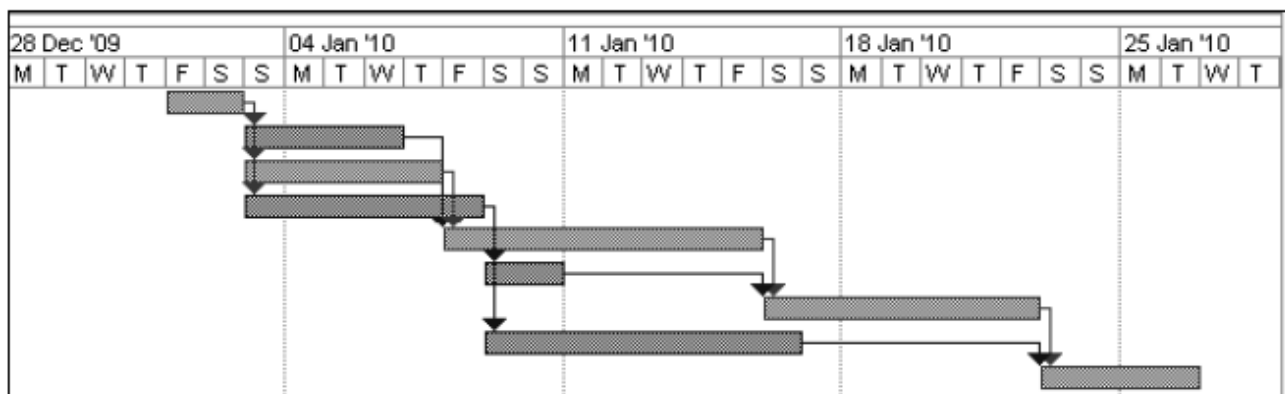


Figure 36: WBS for the Project



b. Time analysis

ID	Task name	Duration (days)	Precedence	Early start	Early finish	Late start	Late finish	Total float
1	A	2	-	01-01-10	02-01-10	01-01-10	02-01-10	0
2	B	4	1	03-01-10	06-01-10	04-01-10	07-01-10	1
3	C	5	1	03-01-10	07-01-10	03-01-10	07-01-10	0
4	D	6	1	03-01-10	08-01-10	08-01-10	13-01-10	5
5	E	8	2,3	08-01-10	15-01-10	08-01-10	15-01-10	0
6	F	2	4	09-01-10	10-01-10	14-01-10	15-01-10	5
7	G	7	5,6	16-01-10	22-01-10	16-01-10	22-01-10	0
8	H	8	4	09-01-10	16-01-10	15-01-10	22-01-10	6
9	I	4	7,8	23-01-10	26-01-10	23-01-10	26-01-10	0

c. Resource graph

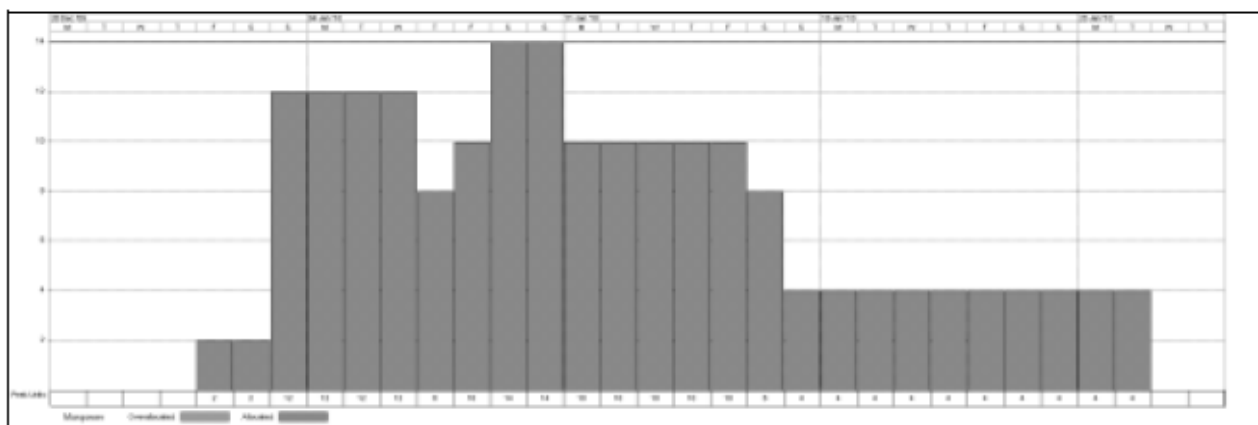


Figure 37: Gantt Chart, Time Analysis and Resource Graph with Unlimited Manpower

For limited manpower (8 no.)

ID	Task Name	Duration	Start	Finish	Predecessors	Total Slack
1	A	2 days	01-01-10	02-01-10		0 days
2	B	4 days	03-01-10	06-01-10	1	3 days
3	C	5 days	03-01-10	07-01-10	1	2 days
4	D	6 days	07-01-10	12-01-10	1	0 days
5	E	8 days	13-01-10	20-01-10	2,3	2 days
6	F	2 days	21-01-10	22-01-10	4	0 days
7	G	7 days	23-01-10	29-01-10	5,6	0 days
8	H	8 days	21-01-10	28-01-10	4	1 day
9	I	4 days	30-01-10	02-02-10	7,8	0 days

Figure 38: WBS with Total Slack



a. Gantt chart

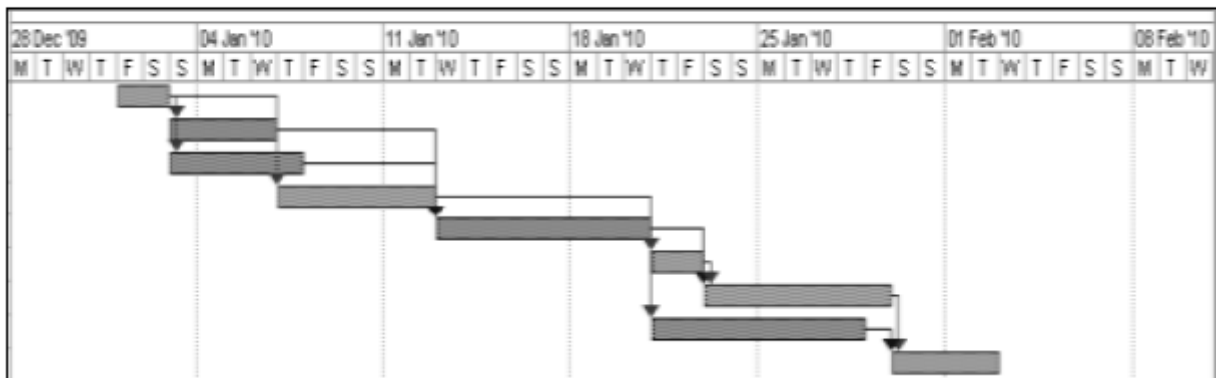


Figure 39: Gantt Chart

b. Time analysis

ID	Task Name	Duration	Predecessors	Early Start	Early Finish	Late Start	Late Finish	Total Slack
1	A	2 days		01-01-10	02-01-10	01-01-10	02-01-10	0 days
2	B	4 days	1	06-01-10	06-01-10	03-01-10	09-01-10	3 days
3	C	5 days	1	05-01-10	07-01-10	03-01-10	09-01-10	2 days
4	D	6 days	1	07-01-10	12-01-10	03-01-10	12-01-10	0 days
5	E	8 days	2,3	15-01-10	20-01-10	08-01-10	22-01-10	2 days
6	F	2 days	4	21-01-10	22-01-10	13-01-10	22-01-10	0 days
7	G	7 days	5,6	23-01-10	29-01-10	23-01-10	29-01-10	0 days
8	H	8 days	4	22-01-10	28-01-10	13-01-10	29-01-10	1 day
9	I	4 days	7,8	30-01-10	02-02-10	30-01-10	02-02-10	0 days

c. Resource graph

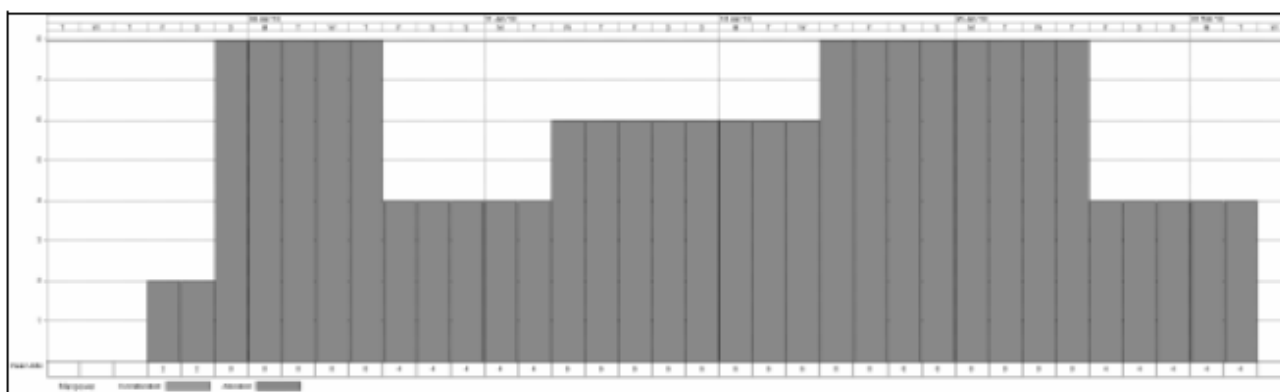


Figure 40: Time Analysis and Resource Graph with Limited Manpower.



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ID	Task Name	Duration	Early Start	Early Finish	Late Start	Late Finish	Total Slack
1	A	2 days	01-01-10	02-01-10	01-01-10	02-01-10	0 days
2	B	4 days	03-01-10	06-01-10	03-01-10	06-01-10	0 days
3	C	5 days	03-01-10	07-01-10	03-01-10	07-01-10	0 days
4	D	6 days	07-01-10	12-01-10	07-01-10	12-01-10	0 days
5	E	8 days	08-01-10	20-01-10	15-01-10	22-01-10	2 days
6	F	2 days	13-01-10	22-01-10	21-01-10	22-01-10	0 days
7	G	7 days	23-01-10	29-01-10	23-01-10	29-01-10	0 days
8	H	8 days	13-01-10	28-01-10	22-01-10	29-01-10	1 day
9	I	4 days	30-01-10	02-02-10	30-01-10	02-02-10	0 days

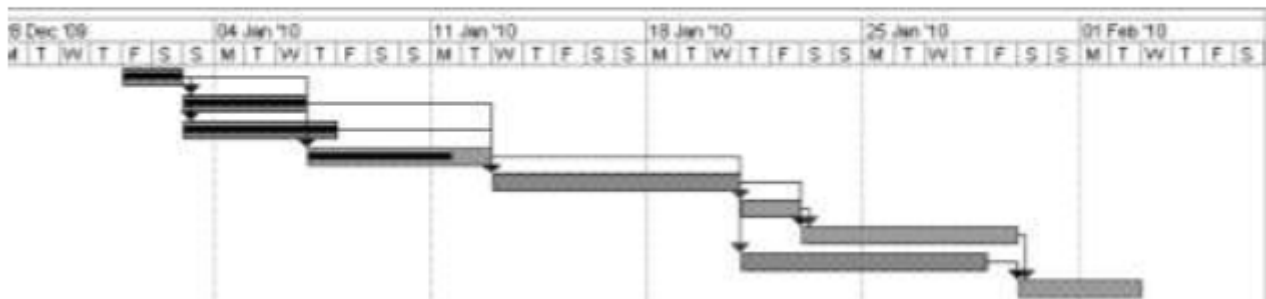
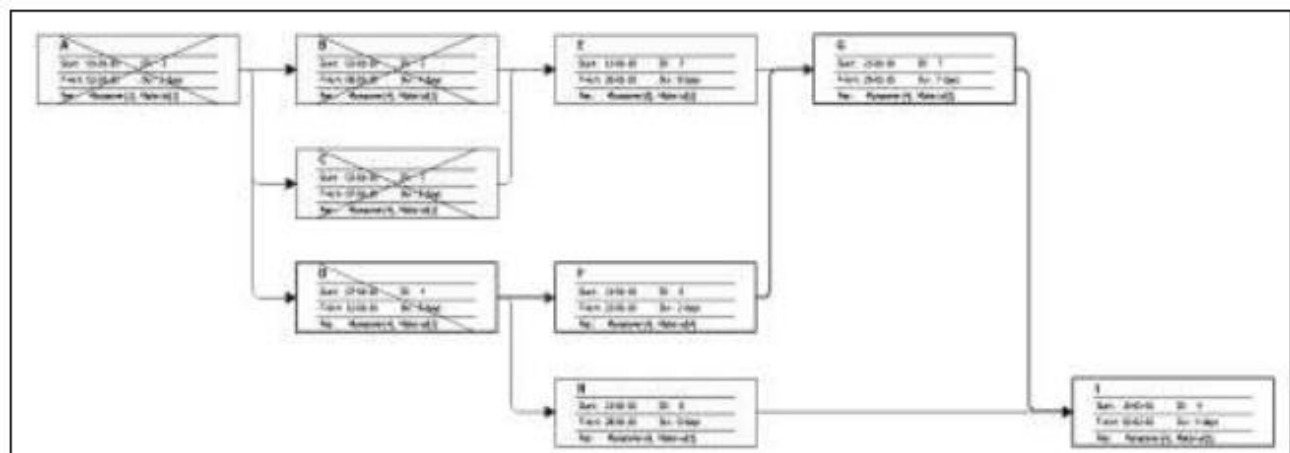


Figure 41: Tracking Gantt Chart



Note: Fully crossed boxes denote the completed tasks and half crossed denote the partially completed tasks

Figure 42: Network Showing the Work Progress



Summary

In this segment, you learned that:

- earlier, project management software was run on mainframe computers and its capability, ease of operation and application level were not satisfactory;
- gradually, more and more projects of varying sizes and complexities emerged which forced the project management software to grow in quantity and quality;
- today, we have a large number of PC-based project management software packages available;
- they are more flexible with improved capability and can handle large projects with speed and accuracy;
- project management software has many advantages including increased speed and accuracy with reduced effort, ability to carry out specialised functions, affordability, ease of use and provision for maintenance of records; and
- all types of project management software share some common features, which include data entry features, graphics, time analysis, resource scheduling, output reports, updating, importing/exporting, project monitoring and tracking.